INCH-POUND

MIL-F-51194C(EA)
6 September 1990
SUPERSEDING
MIL-F-51194B(EA)
3 November 1989

MILITARY SPECIFICATION

FILTER, PARTICULATE, 20 CFM, M19

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of particulate filter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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STANDARDS

MILITARY

MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance - Test Methods
MIL-STD-1168 - Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-19-1175 - Filter, Particulate, 20 CFM, M19 Assembly
136-42-850-2 - Penetrometer, Filter Testing, DOP, Q76, Assembly

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-1175 - Filter, Particulate, M19

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CTF, Rock Island, IL 61229-7300.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-1175 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

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3.1.2 Components. All components of the particulate filter shall conform to the specifications and drawings listed on Drawing 5-19-1175 and subsidiary drawings.

3.2 Assembly. The particulate filter shall be assembled as specified on Drawing 5-19-1175.

3.3 Resistance to airflow. The resistance to airflow shall be no greater than 1.25 inches of water (311 Pa) at an airflow rate of 20 cubic feet per minute (cfm) (9.4 liter per second) when tested as specified in 4.4.6.1.

3.4 Dioctylphthalate (DOP) smoke penetration. The DOP smoke penetration shall be no greater than 0.05 percent at an airflow rate of 20 cfm (9.4 liter per second) when tested as specified in 4.4.6.1.

3.5 Resistance to airflow, DOP smoke penetration, and air leakage after rough handling. The filter shall meet the requirements of 3.3, 3.4, and 3.7 after rough handling for 15 minutes at an amplitude of 3/4 inch (19 millimeters) with the filter face, pleats, and separators in a vertical position, when tested as specified in 4.4.6.2.

3.6 First article. Prior to the start of regular production, a first article sample of particulate filters shall be produced in accordance with this specification for examination and test (see 4.2).

3.7 Air leakage. Air leakage shall be not greater than 15 cubic centimeters per minute when the filter is internally pressurized to a pressure of not less than 12 inches of water, when tested as specified in 4.4.6.3.

3.8 Workmanship. The particulate filter shall be free from chipped, burred, or bent metal sections and loose or damaged gaskets; foreign matter, such as oil or viscous substances, and rips, tears, or holes in the filtering medium.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's

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overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Sample. The first article sample shall consist of 50 particulate filters manufactured using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.2 Inspections to be performed. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

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4.3.3 **Acceptance criteria.** If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the filters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and offered for acceptance at one time; however, no more than one lot of filter material shall be represented in any one lot of finished filters. Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.4.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.4.5 and, when specified, table I. Samples shall be selected at random.

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TABLE I. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.4.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.4.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.4.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the

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item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.4.5 **Classification of characteristics.** Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table I for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(a)	Filter, particulate, 20 CFM, M19		5-19-1175
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	Resistance to airflow	100 percent Acc 0, Rej 1	3.3	4.4.6.1
2	DOP smoke penetration	100 percent Acc 0, Rej 1	3.4	4.4.6.1
Major				
*101	Resistance to rough handling	*	3.5	4.4.6.2**
102	Air leakage	Table I, Level VII	3.7	4.4.6.3
103	Component evident	Table I, Level VII	3.1	VI
104	Component correctly assembled	Table I, Level VII	3.2	VI
105	Marking correct	Table I, Level VII	3.1	VI
106	Overall dimensions correct	Table I, Level VII	3.1	CE
107	Workmanship	Table I, Level VII	3.8	VI
Minor				
201	Marking legible	Table I, Level IX	3.1	VI

NOTES:

*For the first article only, thirteen samples shall be tested: Acc 0, Rej 1

**Destructive test

CE - Commercial inspection equipment

VI - Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(b)	Packaging		P5-19-1175
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Item completely clean and dry prior to unit packing	Table I, Level IX	5.1	VI
102	Cushioning pads evident, correct and properly placed	Table I, Level IX	5.1	VI and CE
103	Unit pack container and closure evident and correct	Table I, Level IX	5.1	VI and CE
104	Marking evident, correct and legible	Table I, Level IX	5.1	VI and CE
105	Unit pack container marking correct and evident	Table I, Level IX	5.1	VI and CE

NOTES:
This paragraph is not applicable to packaging for interplant shipment.
CE - Commercial inspection equipment
VI - Visual inspection

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4.4.6 Tests.

4.4.6.1 Resistance to airflow and DOP smoke penetration. The filters shall be tested for resistance to airflow and DOP smoke penetration with Q76 Filter Testing Penetrometer (Drawing 136-42-850-2).

4.4.6.2 Resistance to rough handling. The filters shall be subjected to rough handling in accordance with method T105.10 of MIL-STD-282, and then tested for resistance to airflow and DOP smoke penetration in accordance with 4.4.6.1, and tested for air leakage in accordance with 4.4.6.3.

4.4.6.3 Air leakage. The sample filters shall be tested for leakage using a test apparatus which shall have two flat (within .005 inch) smooth (32 micro inch) surfaced metal plates at least .25 inch thick that can be clamped to the gaskets on the inlet and outlet of the filter. The clamping mechanism shall be controlled so that the total combined compression of the two gaskets does not exceed .032 inch. There shall be provisions in one of the plates which shall be fixed to the base of the apparatus for pressurized air to be introduced into the filter under test through a flow meter capable of measuring a flow of 15 $\pm$ 1 cubic centimeters per minute of air flow. There shall also be provisions for measuring the pressure inside the filter between these plates through the use of a manometer or pressure gage capable of measuring pressures of 12.00 inches of water with an accuracy of $\pm$ .06 inches of water. The internal pressure measurement shall be taken through a separate passage through one of the two plates from the passage that is used to introduce the measured flow of air. The apparatus shall also have a flow regulation valve by which the air flow can be controlled. The test shall be conducted by clamping the filter to be tested between the two metal plates, opening the air flow control valve, and adjusting the flow until a pressure of at least 12 inches of water is established within the filter. The flow rate that is required to maintain this pressure is the air leakage value. The filter shall be determined to be defective if the air leakage value exceeds 15 cubic centimeters per minute.

5. PACKAGING

5.1 Military packaging. For all shipments, other than interplant shipments, packaging (preservation, unit packing, packing and marking) shall be as specified in SPI P5-19-1175.

5.2 Interplant shipment (see 6.4). Military packaging as specified in 5.1 is not applicable to interplant shipments. The filters shall be packaged to provide adequate protection from physical damage, and marked to identify the filters and provide any other required data during shipment from the source to the first receiving activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 **Intended use.** The particulate filter covered by this specification is intended for use in the M13 and M13A1 filter units.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)
(c) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(d) Whether interplant shipment is required
(e) Level of packing required
(f) Shelf life codes

6.3 **Submission of alternative inspection provisions.** Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.4 **Interplant shipment.** Packaging and marking for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are from a vendor to a subcontractor or a prime contractor, or between contractors and subcontractors, or from a vendor or contractor to a military arsenal, plant or other activity for evaluation, immediate use, or further processing as specified in the applicable contract.

6.5 **Subject term (key word) listing.**

Airflow resistance
Dioctylphthalate (DOP)
Filter, gas

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Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A001

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51194C(EA)	FILTER, PARTICULATE, 20 CFM, M19

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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